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A scalable infrastructure model for carbon capture and storage: *SimCCS*

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ABSTRACT

In the carbon capture and storage (CCS) process, CO_2 sources and geologic reservoirs may be widely spatially dispersed and need to be connected through a dedicated CO_2 pipeline network. We introduce a scalable infrastructure model for CCS (simCCS) that generates a fully integrated, cost-minimizing CCS system. SimCCS determines where and how much CO_2 to capture and store, and where to build and connect pipelines of different sizes, in order to minimize the combined annualized costs of sequestering a given amount of CO_2. SimCCS is able to aggregate CO_2 flows between sources and reservoirs into trunk pipelines that take advantage of economies of scale. Pipeline construction costs take into account factors including topography and social impacts. SimCCS can be used to calculate the scale of CCS deployment (local, regional, national). SimCCS' deployment of a realistic, capacitated pipeline network is a major advancement for planning CCS infrastructure. We demonstrate simCCS using a set of 37 CO_2 sources and 14 reservoirs for California. The results highlight the importance of systematic planning for CCS infrastructure by examining the sensitivity of CCS infrastructure, as optimized by simCCS, to varying CO_2 targets. We finish by identifying critical future research areas for CCS infrastructure.

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1. Introduction

Carbon capture and storage (CCS) is an essential component of the portfolio of technologies for reducing CO_2 emissions (Anderson and Newell, 2004; IPCC, 2005). CCS involves capturing and compressing CO_2 at fixed point sources (such as coal and natural gas power plants, oil refineries, and ethanol plants), transporting the liquid or supercritical CO_2 through dedicated pipelines, and injecting and storing the CO_2 in geologic reservoirs (such as depleted oil and gas fields, saline aquifers, and deep-sea sediments). The goal of CCS is to store CO_2 away from the atmosphere for hundreds or thousands of years without fundamentally restructuring energy infrastructure in the short term. CCS is an important climate mitigation technology because, unlike other technologies, it can be integrated with the organization of CO_2-emitting industries that make up the present economy. Because it utilizes mature technologies from the oil and gas industry, CCS can be deployed both in the near term and throughout the 21st century (Pacala and Socolow, 2004; Stephens and van der Zwaan, 2005). Consequently, CCS is often viewed as a bridging technology in that it enables low cost and readily available fossil fuels to be consumed as society progresses to alternative energy sources in the long term (Riahi et al., 2003; IPCC, 2005). Further, implementing CCS technology, particularly the costly capture and compression process, adds costs to operations that emit CO_2 and thus increases fossil fuel energy prices, making other energy sources more cost competitive.

The infrastructure supporting CCS will have to be deployed at a considerable scale in order to make a significant contribution to climate mitigation; a typical 1 GW coal-fired power plant produces approximately 6 $MtCO_2$/yr. CO_2-emitting sectors of the US economy (e.g., coal-fired power, cement manufacturers, oil refineries) emitted 1000 times this amount, approximately 6 $GtCO_2$, in 2005, (EIA, 2007). According to MIT (2007), if all of the CO_2 produced in 2005 by coal-fired power plants in the US were to be transported and injected for storage, the amount of CO_2 would be "equivalent to three times the weight, and under typical operating conditions, one-third of the annual volume of natural gas transported by the US gas pipeline system… if 60% of the CO_2 produced from US coal-based power generation were to be captured and compressed to a liquid for geologic sequestration, its volume would about equal the total US oil consumption of 20 million barrels per day," (p. ix). Clearly, infrastructure at this massive scale requires detailed and systematic forward planning. Consequently, it is essential to be able to minimize the costs of CCS infrastructure needed to meet CCS targets.

A comprehensive CCS infrastructure model should *simultaneously* consider and integrate seven key decisions: (1) how much CO_2 to capture (2) at which sources; (3) where to construct pipelines and (4) of what size; (5) which reservoirs should store

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CO_2 and (6) how much to inject; and (7) how to distribute CO_2 from the dispersed sources through the network to the reservoirs. These core decisions have to be simultaneously optimized in order to understand the cost of the integrated CCS process. Identifying where to construct a realistic pipeline network, the only feasible transportation mode for large volumes of CO_2 (Svennson et al., 2004; Doctor et al., 2005), has provided a great challenge for previous infrastructure models and has always been highly approximated or completely disregarded. Moreover, the high spatial dependency between CCS components (CO_2 sources, reservoirs, and pipelines) makes this simultaneous optimization a very complex and challenging problem. As a result, previous work has focused on one or two of these seven decisions and relied on assumptions, particularly for the pipeline network, to make their models tractable. For example, models typically assume: that a single CO_2 pipeline directly connects a single source to a single injection site (MIT, 2006); that these pipelines will be straight (ISGS, 2005); that there is a minimum and maximum distance for pipelines between sources and reservoirs (Dooley et al., 2006); that all CO_2 at a source must be captured regardless of system-wide economics (ISGS, 2005; Dooley et al., 2006); or that sources and reservoirs be matched by incrementally branching out to that which is closest (Kobos, et al., 2007). None of these CCS infrastructure models deploy a realistic network with capacitated pipelines or can generate high-capacity trunklines.

In this paper, we introduce a *scalable infrastructure model for CCS* (*simCCS*). *SimCCS* is a geospatial-optimization, economic–engineering model that simultaneously and optimally considers all components of CCS infrastructure to identify both the cost and spatial arrangement of an integrated CCS system. *SimCCS* is supported by an algorithm to identify candidate pipeline segments (i.e., routes one might want to consider as parts of the final optimal network) in a way that minimizes construction costs including the impact on environmentally sensitive areas and the higher costs of building through cities, of crossing water, roads, and railroads, and of building through difficult topography. The model allows for partial amounts of CO_2 to be captured from CO_2 sources, approximating the lower cost to capture a reduced proportion of CO_2 from exhaust gases. *SimCCS* can be used to calculate the cost of CCS infrastructure to meet sequestration targets as well as identify the appropriate scale of CCS deployment (local, regional, national) and estimate carbon tax/permit pricing. *SimCCS* optimization can be used to decide where to retrofit *existing* CO_2 sources with capture technology, or, with a minor modification, to decide where *future* CO_2-emitting facilities should be located. We demonstrate *simCCS* using a set of CO_2 sources (power plants, oil refineries, and cement manufacturers) and CO_2 reservoirs (depleted oil and gas fields) for the state of California and highlight the importance of systematic planning for CCS infrastructure by examining the sensitivity of *simCCS* to variable CO_2 targets and costs. We finish by identifying critical future research areas for CCS infrastructure, in particular how to deploy a CCS distribution system in a robust and resilient manner.

2. Dedicated CO_2 pipeline network

The most significant and novel contribution of, and greatest challenge for, *simCCS* is constructing a realistic, efficient pipeline network connecting optimized CO_2 sources and reservoirs. *SimCCS* forms the network by optimally selecting arcs (as well as pipeline capacities) from a candidate pipeline network—a large set of potential routes to be considered as part of the optimal final network. Currently, these candidate network data, and a research methodology to construct these data, does not exist. This absence is in part responsible for CCS infrastructure models simplifying or ignoring the pipeline network. A completely integrated CCS system, however, relies on a realistic pipeline distribution system.

SimCCS generates a candidate pipeline network using a pre-optimization algorithm that extracts a vector-based network (i.e., a connected set of nodes and arcs) from a raster-based (i.e., grid cells) "cost" surface. The cost surface used in this paper is a 1-km raster grid covering the contiguous US (Fig. 1), and represents the estimated cost to construct a pipeline across each grid cell (MIT, 2006). Natural gas pipelines have been suggested as an analog for estimating the cost of constructing CO_2 pipelines (e.g., McCoy and Rubin, 2008). The cost grid was generated by reverse-engineering the contribution of geographical features (including slope, urban areas, national/state parks, and road/rail/river crossings) to the cost of natural gas pipeline construction in the US (MIT, 2006). Each of these geographical features is represented by a separate layer of grid cells (Fig. 1), where the value in each cell is a weight representing how much that feature multiplies the base cost of pipeline construction. Using simple map algebra (i.e., the order of each feature layer does not matter), the values in each grid layer are summed to obtain the overall construction cost multiplier for each grid cell. Consequently, for instance, the cost to construct a 12-inch pipeline across a 1-km grid cell is calculated by multiplying the per-kilometer base cost of a 12-inch CO_2 pipeline by the weight in that cell. The cost to construct a single pipeline between any two points on the cost

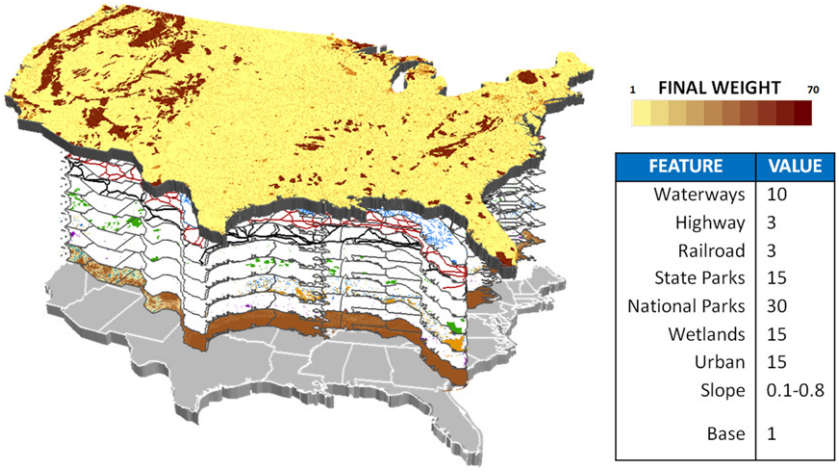


Fig. 1. Cost surface (modified from MIT, 2006).

surface can be calculated by employing Dijkstra's (1959) shortest path algorithm. In this implementation, the distance to cross any 1-km grid cell follows orthogonal or diagonal directions via the center point of the cell.

Using a modified Dijkstra's algorithm on the weighted-cost surface, the *simCCS* pre-optimization algorithm first calculates the lowest-cost paths for direct pipelines between *all* possible node pairs of CO_2 sources and reservoirs. These are individual, unconnected paths not yet linked together into a network, and many portions of these paths are coincident because they follow troughs in the cost surface. Next the algorithm converts the overlapping individual source-to-reservoir routes of adjacent raster cells into a vector-based candidate network of nodes and arcs. To do so, it divides each path of cells into a series of arcs and nodes, where arcs represent a potential segment of pipeline and nodes represent junctions and endpoints of arcs. After that, the algorithm removes overlapping arcs (with the same start and end nodes) where many individual source-to-reservoir pipelines had overlapped. The algorithm can then trim this candidate network further by removing superfluous arcs and grid artifacts such as where nodes are in close proximity.

3. SimCCS

SimCCS constructs an integrated CCS system using a mixed integer linear programming (MILP) model. MILP approaches are ideally suited for complex optimization problems with multiple and competing decisions and constraints (Bertsimas and Tsitsiklis, 1997), as is the case with *simCCS*. MILP models replicate complex, nonlinear processes by breaking them apart into constituent linear processes. *SimCCS*' primary objective is to *minimize* total cost, including fixed infrastructure costs and operating costs, given a target CO_2 capture/storage amount. The versatility of MILP, however, would also allow *simCCS* to seamlessly *maximize* the amount of CO_2 captured given a fixed budget. The *simCCS* model can be solved using a general MILP solver. In this paper, we use ILOG's CPLEX 11.0, a state-of-the-art mixed integer optimizer.

The *simCCS* model is a MILP because it combines continuous (linear) decision variables that can take on any value greater or equal to zero and integer (in this case binary) decision variables that can take on values of 0 or 1. The continuous variables represent how much CO_2 to: capture (a_i) at any node i; transport (x_{ij}) from any i to any j; or inject (b_j) at any node j. The binary variables represent decisions whether or not to build: a capture/compression facility at any source node i (s_i); a pipeline of diameter d from i to j (y_{ijd}); or injection wells at reservoir node j (r_j). The subscript d on the pipeline construction variables is a key element of the model for incorporating economies of scale, in that it allows the model to build pipelines of the most cost-efficient diameter for transporting any particular amount of CO_2. In our network-based approach, the amount of CO_2 to flow through any pipeline arc is not known in advance because it depends on which sources and reservoirs are opened and which flows can be efficiently combined into trunklines.

The *simCCS* model formulation, like any MILP problem, consists of an objective function (minimize total costs) and a set of constraints that have to be satisfied. The objective function (1) minimizes the sum of fixed and variable costs. Fixed costs for sources, pipelines, and reservoirs approximate the costs to install capture technology; purchase land and construct pipelines; and select, drill, and equip injection sites. Variable costs represent variable CO_2 capture costs (i.e., per tonne of CO_2), maintenance/pumping through the pipeline network, and the operating costs of the injection process. Constraints (2) and (3), respectively, ensure that CO_2 flow through pipelines is less than the maximum capacity and greater than the minimum capacity. Constraint (4) is a mass balance constraint stipulating that all CO_2 captured at or flowing into a node must be injected or transported out of the node. Constraints (5) and (6) set the maximum CO_2 supply and storage capacity for each source and reservoir. Constraint (7) stipulates the target system-wide CO_2 capture amount, while constraint (8) allows only one pipeline (of any size) to be constructed on any potential arc. Constraints (9)–(11) specify the zero-one nature of decision variables; for instance, one either entirely constructs a pipeline ($y_{ijd}=1$) or not at all ($y_{ijd}=0$). Constraints (12)–(14) force all CO_2 quantities in the model to be non-negative.

$$\text{MINIMIZE} \quad \sum_{i\in S}(F_i^s s_i + V_i^s a_i) + \sum_i \sum_{j\in N_i} \sum_d F_{ijd}^p y_{ijd} + \sum_i \sum_{j\in N_i} V_{ij}^p x_{ij} + \sum_{j\in R}(F_j^r r_j + V_j^r b_j) \tag{1}$$

$$\text{S.T.} \quad x_{ij} - \sum_d \max Q_{ijd}^p y_{ijd} \leqslant 0 \quad \forall i,j \in N_i \tag{2}$$

$$x_{ij} - \sum_d \min Q_{ijd}^p y_{ijd} \geqslant 0 \quad \forall i,j \in N_i \tag{3}$$

$$\sum_{j\in N_i} x_{ij} - \sum_{j\in N_{i\notin R}} x_{ji} - a_i + b_i = 0 \quad \forall i \tag{4}$$

$$a_i - Q_i^s s_i \leqslant 0 \quad \forall i \in S \tag{5}$$

$$b_j - Q_j^r r_j \leqslant 0 \quad \forall j \in R \tag{6}$$

$$\sum_{i\in S} a_i \geqslant T \tag{7}$$

$$\sum_d y_{ijd} \leqslant 1 \quad \forall i,j \in N_i \tag{8}$$

$$y_{ijd} \in 0,1 \quad \forall i,j \in N_i, d \in D \tag{9}$$

$$s_i \in 0,1 \quad \forall i \in S \tag{10}$$

$$r_j \in 0,1 \quad \forall j \in R \tag{11}$$

$$x_{ij} \geqslant 0 \quad \forall i,j \in N_i \tag{12}$$

$$a_i \geqslant 0 \quad \forall j \in S \tag{13}$$

$$b_j \geqslant 0 \quad \forall j \in R \tag{14}$$

Decision variables:

x_{ij} units of CO_2 transported from node i to node j (tonnes)

y_{ijd} $\begin{cases} 1, & \text{if a pipeline is constructed from node } i \text{ to node } j \text{ with diameter } d \\ 0, & \text{otherwise} \end{cases}$

s_i $\begin{cases} 1, & \text{if source at node } i \text{ is opened} \\ 0, & \text{otherwise} \end{cases}$

r_j $\begin{cases} 1, & \text{if reservoir at node } j \text{ is opened} \\ 0, & \text{otherwise} \end{cases}$

a_i amount of CO_2 produced at node i (tonnes)

b_j amount of CO_2 sequestered/stored at node j (tonnes)

Inputs:

F^s, F^p, F^r	fixed cost for opening a source, constructing a pipeline, or opening a reservoir ($)
V^s, V^p, V^r	variable cost for capturing CO_2 from a source, transport through a pipeline, or into a reservoir ($ per tonne)
Q^s, Q^p, Q^r	CO_2 capacity of a source node, pipeline, or reservoir (tonnes)
T	target amount of CO_2 to be sequestered (tonnes)

Sets:

N_i, N_j	nodes adjacent to nodes i or j
R	reservoir nodes
S	source nodes
D	pipeline diameters

Because the storage capacity of geologic reservoirs is given as a maximum total volume rather than as an annual limit, everything else in the model, such as capture, flow, and injection variables and capacities, have been converted to a lifetime basis (25 years in this case study) for internal consistency. Of course, if reservoir storage capacity could be expressed as an annual injection capacity, the entire model could be redefined on an annual basis with appropriate changes to definitions and coefficients but with no change in the fundamental mathematical structure.

4. Estimated costs and capacities

The costs incurred at the sources, reservoirs, and in the transportation network are estimated as described below and in the Appendix. Because *simCCS* does not have a temporal component, all costs must be in their present values for comparison. For fixed costs, the financing is removed from published estimates of the levelized annual investment costs to convert them to their present total lump-sum values. Variable costs, on the other hand, are based on discounted present values for each year over the lifetime of the system (25 years in this case study).[1]

CO_2 source capture costs are derived from the literature cited in IPCC (2005). Published studies typically include the cost of compression to a somewhat arbitrarily determined pressure for CO_2 transportation by pipeline. Compression is included in capture cost studies in order to compensate for processes that operate, and thus capture CO_2, at different initial pressures. The final pressure, however, is not uniform across previous studies so the compression costs are normalized, using the assumptions in the original study, for compression to 10 MPa. Pressurization requirements above 10 MPa are based on an estimated distance that the source's CO_2 will travel through a pipeline specifically sized for that source's quantity of CO_2. These pressurization requirements are calculated by basic representations of fluid flow through pipelines and the resulting pressure drop over the pipeline's estimated distance in Eqs. 15 and 16 (Appendix). *SimCCS* does not simultaneously optimize and iterate pressure drops so all CO_2 pressurization is assumed to occur at the source. These simplifying assumptions are necessary in order to utilize existing published cost data, but the model structure is flexible enough to be able to include the fixed and variable costs for pressurization/pumping stations in the costs of the candidate pipeline arcs rather than the sources (see future work in Section 7).

Consistent with most capture cost studies, it is assumed that as much as 90% of the total amount of CO_2 produced by the source can be captured. Based on this maximum limit amount of CO_2 that can be captured from each source, *simCCS* chooses how much should be captured optimally, based on the overall costs of the integrated CCS system.

Reservoir costs are also modeled with fixed and variable costs, converted to present values. The fixed reservoir cost includes the up-front costs of site evaluation and screening, well drilling costs, and repressurization equipment costs, if necessary. Variable cost per tonne injected depends on the flow of CO_2 to the reservoir and the geologic properties of the reservoir. This flow rate is spread out over multiple injection wells according to the "injectivity" of the reservoir.[2] Reservoir injectivity is calculated based on an integration of the Darcy flow equation for fluid flow in a porous media and is determined by the geologic parameters of the reservoir (formation permeability, reservoir thickness, etc.) in Eq. (17) (Appendix) and the properties of the injected fluid.[3] Reservoir temperature and pressure determine the density and viscosity of the CO_2 in the subsurface, which are calculated from regressions describing the equations of state for CO_2 (Span and Wagner, 1996) in Eqs. (19) and (20) (Appendix). These CO_2 properties also determine the reservoir injectivity. Site evaluation and screening costs are taken from McCoy and Rubin (2005). Well drilling costs, in Eqs. (21)–(23) (Appendix), are based on analyses undertaken by Advanced Resources International (e.g., ARI, 2005) and an analysis of data presented by the Joint Association Survey of Drilling Costs (API, 2000). It is assumed that the CO_2 arrives at the injection reservoir at 10 MPa, and further repressurization, if necessary, is conducted at the reservoir. The number of necessary injection wells and CO_2 repressurization (if necessary) is based on iterations of reservoir injection pressure (P_2 in Eq. (17)), injectivity (Eq. (17)), CO_2 gravitational head gain (19), and pressure drop (Eq. (16)) in the three-inch diameter injection wells. The estimated CO_2 capacity of each reservoir is based on the density of CO_2 in the subsurface as it would replace the volume of fluid, such as oil, which has been produced from the reservoir.

It is assumed that the minimum pressure is 10 MPa, in order to keep the CO_2 above its critical pressure of 7.4 MPa. All pump and compressor costs for the sources and reservoirs are taken from IEAGHG (2002, 2006), with efficiencies set at 75%, electricity priced at $70/MWh, and a cost of oil of $60/bbl.

Pipeline construction costs are estimated from 15 years of pipeline construction in the US (1990–2005) as published in the *Oil & Gas Journal*. The regression uses a Cobb–Douglas function to estimate the nonlinear relationship of construction cost on pipeline diameter and length. For the purpose of *simCCS*, pipeline construction costs were normalized as a constant amount per unit length, for a 100 km pipeline, for each pipeline diameter. Annual pipeline operating costs are taken from Heddle et al., (2003), McCoy and Rubin (2005), MIT (2006), and McCoy and Rubin (2008). Pipeline costs exhibit tremendous economies of utilization and economies of scale.

Fig. 2 shows the pipeline unit-cost-per-tonne function implied by the fixed costs, variable costs, and capacities of each diameter pipeline in the model. Economies of full utilization can be seen in the concave asymptotic L-shape of the curve for each individual pipeline size as the fixed costs of construction are spread over larger and larger quantities up to the pipeline's capacity. Economies of scale can be

[1] Financing is removed using the assumptions in the original study, and then the present total lump-sum values are converted to levelized annual investment costs using an 11% fixed charge factor.

[2] This approach to "injectivity" determines the number of injection wells that are needed, but not their special arrangement. *SimCCS* is scaleable to different resolutions and, as such, could be used to optimize the layout of an injection reservoir where a fluid is being produced, such as oil for CO_2-EOR or brine with CO_2 storage, if the spatial profile of the reservoir injection pressure were incorporated.

[3] This formulation is consistent with Law and Bachu (1996), which is used by McCoy and Rubin (2005).

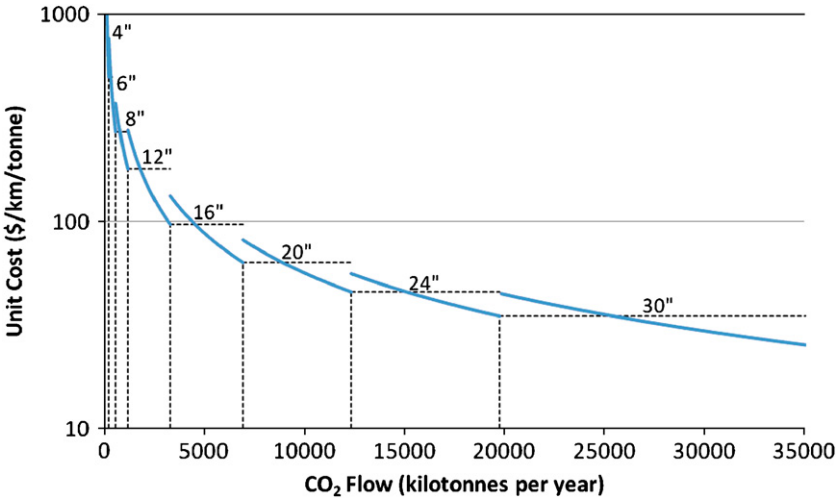


Fig. 2. Pipeline costs (dollars per km per tonne of CO_2) for 10 pipeline capacities.

Table 1
Estimated costs and capacities for California data set

	CO_2 capacity ((*S*) kt/yr; (*R*) kt)		Estimated costs ($/t$CO_2$)	
	Mean	Standard deviation	Mean	Standard deviation
37 sources	1752	986	44.17	7.12
14 reservoirs	110,639	21,911	3.60	2.05

seen in the lower envelope of all 10 curves and the tremendous decline in unit costs for larger and larger diameters. This type of cost curve combined with network-design approach of *simCCS* enables the model to achieve major cost savings vis-à-vis models that only consider direct pipelines between sources and reservoirs.

5. California study

Here we demonstrate *simCCS* using the 37 largest CO_2 sources (21 natural gas power plants, 1 coal power plant, 10 oil refineries, and 5 cement manufacturers) and 14 reservoirs (depleted oil fields). These sources and reservoirs are the sets from which *simCCS* chooses to deploy the integrated CCS system. Depending on the interaction and spatial distribution of costs, capacities, and scale (the target amount of CO_2 to be stored) only some subset of these sources and reservoirs might actually be chosen for deployment. Reservoir selection was limited to depleted oil fields because their CO_2 storage potential is well understood and characterized, allowing reasonable engineering and economic estimates to be made. CO_2 source data were taken from WestCarb (2008), and accurate location information generated using address matching (many original source locations are incorrect or aggregated to county centroids). Reservoir data were sourced from WestCarb (2008) with each reservoir collapsed to a single node defined by its interior centroid. Table 1 highlights capacities and costs for sources and reservoirs in this study. The pipeline costs were artificially deflated for this study to emphasize *simCCS*' capability to construct an integrated pipeline network.

6. Results

Fig. 3 shows the potential routes determined by the *simCCS* pre-optimization algorithm. It is evident that the algorithm is choosing potential routes that avoid the darker patches (higher

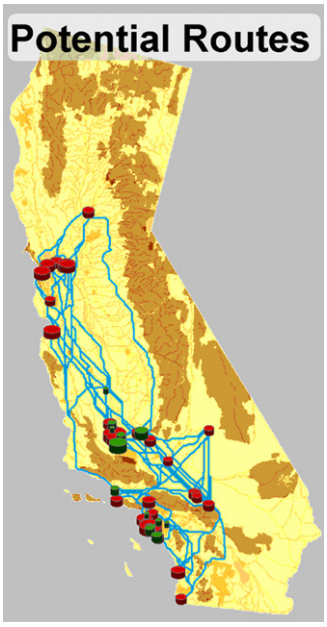


Fig. 3. Potential pipeline routes (blue) between 37 largest CO_2 sources (red) and 14 largest CO_2 capacity oil fields (green).

cost areas) in the cost surface. These routes connect every combination of the 37 sources and 14 reservoirs with the potential to be deployed by the process described in Section 2. Note that it is important to generate candidate arcs that give the model the option of connecting sources to other sources, and reservoirs to other reservoirs, as one of the possible ways for the model to combine shipments into trunklines.

Fig. 4 presents the optimized deployment of the integrated CCS system—chosen from the potential routes, 37 source nodes, and 14 reservoir nodes—at 3 different scales: 10, 30, and 50 $MtCO_2$/yr. The least-expensive way to capture and store 10 $MtCO_2$/yr, according to *simCCS*, is to build 258 km of a mostly linear (there is one small spur), multistep east–west pipeline network across south central California. The model captures CO_2 at 5 sources and injects it into 2 reservoirs, at an average cost per tonne of $41.71.[4]

[4] All cost estimates reported in the results are based on an 11% fixed charge factor.

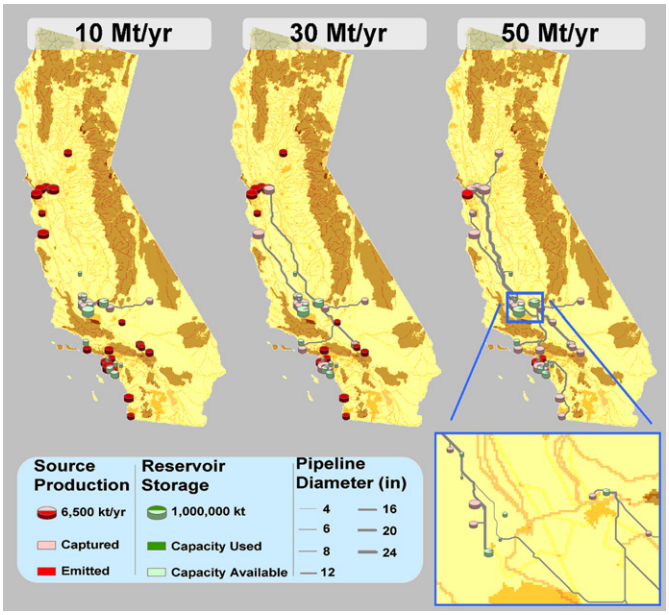


Fig. 4. Optimized deployment of integrated CCS system for 37 largest CO_2 sources and 14 largest CO_2 capacity oil fields in California.

For 30 $MtCO_2$/yr, the model constructs a very different north-south tree network in the central valley. It also opens 18 sources and 8 reservoirs, which collect and store CO_2 from several large nearby sources. To further utilize this reservoir capacity, the model constructs a parallel north-south pipeline from the San Francisco Bay Area. Total cost for this system increases to $43.27/tonne.

Finally, the third map shows an optimal solution for sequestering 50 $MtCO_2$/yr. To reach this high target the model captures CO_2 at all but four of the 37 sources, and injects it into 12 of the 14 reservoirs. Because there are no candidate reservoirs in northern California in this case study, the model builds a high-capacity, 24-inch trunkline from northern to southern California and an elaborate network of spurs and connectors. There is a surprisingly high degree of connectivity in the 50 $MtCO_2$/yr system: there are only 5 separate subnetworks. In fact, 4 of the 5 subnetworks are in southern California, and the LA area alone has 3 separate subnetworks in part because routing through the urban terrain is quite costly. The model develops this strategy in order to utilize all the source capacity that is opened. Since the pipelines are relatively cheap to build compared with the capture and storage facilities, it makes sense to build spurs and connectors to allow the overflow of CO_2 to be redirected to another reservoir, thus storing every last tonne that is captured. For scales above 100 $ktCO_2$/yr, the average source capacity utilization above is 99%, whereas the average reservoir capacity utilization is 3.1%. This discrepancy is due to the fact that source costs are much greater than reservoir costs and reservoir storage capacity is much greater than source production capacity.

These three sample solutions are presented here not as policy recommendations but to illustrate the kinds of networks that *simCCS* is able to construct to take advantage of economies of scale and utilization in pipeline transport, and to fully utilize the capacity created by the large investments in capture/compression technology at sources and the injection capacity at reservoirs. These diagrams also illustrate the great difficulty of optimizing such a system without the aid of MILP and its ability to perform combinatorial optimization of 0–1 investment decisions.

Fig. 5 presents the levelized annual cost curves for cost-minimizing deployment of 37 sources and 14 reservoirs up to the maximum amount of CO_2 that can be captured if all sources are fully utilized (roughly 56 $MtCO_2$/yr). The 3 optimal solutions shown in Fig. 4 correspond to the values of 10,000, 30,000, and 50,000 $ktCO_2$/yr on the *X*-axis.

The total costs are the sum of the source, reservoir, and transportation network costs. As described in Section 4, the source costs are derived from the literature and are thus consistent with previously published studies at $35–55/tonne, depending on financing assumptions. Reservoir costs are consistent with previous estimations of $1.50–5.50/tonne for reservoir costs (e.g., McCoy and Rubin, 2005). (See Table 1 for the costs used in this example.) The transportation network costs of less than $1/tonne for all but the smallest tonnage targets may appear to be at the low end of typical ranges for estimated transportation costs. The underlying data, however, are consistent with previous analyses. The transportation network costs are lower because *simCCS* incorporates, and takes advantage of, increasing returns to scale when CO_2 flows are aggregated into larger diameter pipelines, as well as increasing returns to utilization by spreading the fixed cost of any particular pipeline size over as many tonnes as possible. Also, pipeline construction costs were artificiality reduced for this scenario to emphasize *simCCS*' pipeline network capability. Previous analyses are for single pipelines, albeit many of them, directly connecting sources and reservoirs.

Fig. 5 illustrates that *simCCS* does far more than just minimize the pipeline network costs. Because capture and compression costs make up the lion's share of total costs (around 90% for our study), the model opens the cheapest sources first. Unit capture costs are lowest at 800 $ktCO_2$/yr when the CO_2 from the cheapest source in the system (a coal-fired power plant in San Bernardino, CA) is fully utilized at a capture cost of about $36/tonne. For larger targets, up to 40 $MtCO_2$/yr, the model opens a number of sources that, when fully utilized, average below $40/tonne for capturing. Capture costs only surpass $40/tonne for system targets above 40 $MtCO_2$/yr, after which capture costs climb steeply because of the deployment of the most costly sources. The domination of capture costs in the overall cost picture helps explain the long north–south pipeline that is included in the optimal 30 $MtCO_2$/yr system in Fig. 4. The Bay Area source included in this solution has a capture cost of only $39/tonne when fully utilized, substantially cheaper than some of the unused sources in southern California that are much closer to the reservoirs.

Compared with the capture cost curve, the reservoir cost curve follows a decidedly different pattern, rising sharply between the 10 and 20 Mt/yr systems and then increasing at a more gradual pace beyond that. Meanwhile, the network cost curve shows an average below $0.50/tonne for quantities of 10–20 $MtCO_2$/yr, rising gradually to about $1/tonne to connect all sources to reservoirs. Notice that at 13 and 19 $MtCO_2$/yr the network costs jump upward, as it becomes necessary to build pipelines that are underutilized. This occasionally can happen in such models because the pipelines come in discrete sizes.

Fig. 5 also includes results where *simCCS*' pipeline network (with trunklines) is replaced by pipelines that directly connect each to source to each reservoir; all previous CCS infrastructure models use such direct CO_2 pipelines. The results using direct pipelines are denoted by hollow symbols in Fig. 5. The cost and routes for the direct pipelines are calculated during the first stage of the *simCCS* pre-optimization algorithm. Solutions using direct pipelines can never be an improvement over *simCCS*' networked pipeline system since the direct pipeline paths are contained within the *simCCS* candidate pipeline network. Up to 2 $MtCO_2$/yr, the total CCS cost for the networked and direct pipelines are the same because all components of the solution (i.e., sources, pipelines, reservoirs) are identical. At higher sequestration targets, the total costs using dedicated pipelines gradually increase above

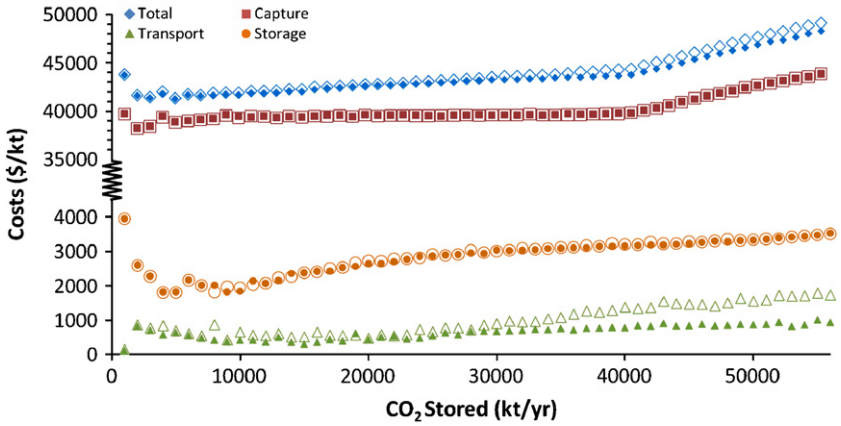


Fig. 5. Average cost curve(s) for *simCCS* deployment of 37 largest CO_2 sources and 14 largest CO_2 capacity oil fields in California—hollow symbols denote use of direct pipelines, where each pipeline *directly* connects a single source with a single reservoir.

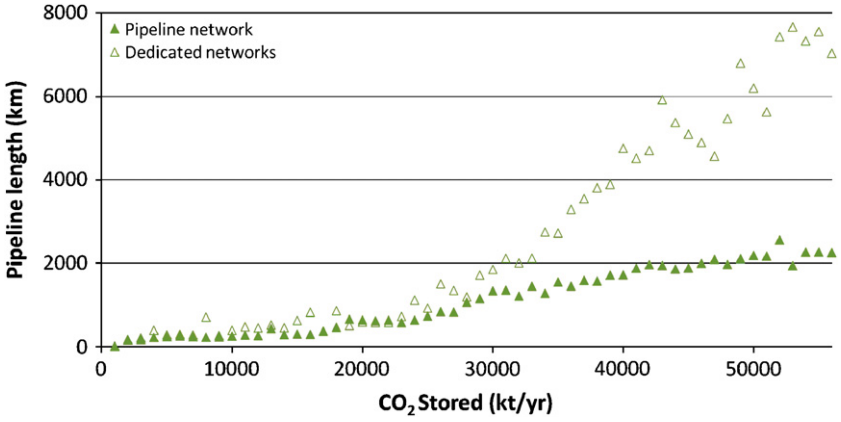


Fig. 6. Total pipeline length using a pipeline network (*simCCS*) and using dedicated pipelines between each source and reservoir.

those for networked pipelines, typically increasing total CCS costs by around 1.5%; for the 54 Mt/yr solution this equates to $48 MM.

The cost of just the pipelines increases by as much as 105% (53 Mt/yr), and is typically around 80%, if direct pipelines are used. For higher sequestration targets, this equates to around $0.80/tonne of CO_2, approximately the same amount that total CCS infrastructure costs are raised. The total length of pipelines built (Fig. 6) increases rapidly if using direct pipelines. For example, if all available CO_2 for the study area is captured (56 Mt/yr), the direct pipelines have a total length of over 7000 km, more than 3 times than the networked pipeline solution. For lower capture amounts ($\leqslant$23 Mt/yr) the direct pipeline solutions can compensate, both in terms of cost and length for, by directly linking low cost and close-by sources and reservoirs. It should be noted, though, that this compensation changes the spatial configuration of which sources and reservoirs are selected in the model. As the target amount increases ($>$23 Mt/yr) the direct pipelines can no longer join close, cheap sources and reservoirs, and the network costs and length escalates.

7. Conclusions

We have introduced a comprehensive CCS infrastructure model, *simCCS*, which estimates CCS costs and solves for optimal spatial deployment of infrastructure. *SimCCS*' most significant contribution to CCS research and technology is the ability to generate a realistic and efficient pipeline network to connect spatially dispersed CO_2 sources and reservoirs. Consequently, for the first time, it is possible to estimate costs for a complete and integrated CCS system as well as determine where and how CO_2 should be captured, transported, and stored. With only minor modification, *simCCS* can be employed to locate where new CO_2 sources should be located (such as power plants) in addition to identifying where existing facilities could be retrofitted. *SimCCS* is uniquely able to systematically address many planning issues for CCS infrastructure and deployment.

We have demonstrated *simCCS* using 37 CO_2 sources and 14 reservoirs for the state of California. The optimal networks in Fig. 4 and the cost trends in Fig. 5 are, of course, not generalizable beyond this case study. They do, however, illustrate the kinds of savings that the model is capable of producing, as well as the strong influence of both geographic variations in pipeline construction costs and differences in capture and injection costs from site to site. Results will vary depending on the relative locations, costs, and capacities of sources and reservoirs, the costs and performance of different technologies, and other application-specific factors. Given that the costs in this study are based on the best data available in the literature, it appears that the greatest savings identified by the model are not the economies of scale in the pipeline construction per se but in constructing a well-connected pipeline *network* that is able to redirect the overflow of CO_2 that results when source and reservoir capacities do not match perfectly. Routing CO_2 through high-capacity trunklines allows *simCCS* to achieve lower pipeline system costs than previously published, but it appears that even greater savings

may be achieved by taking advantage of cheaper sources and reservoirs and by fully utilizing the source capacities and, to a lesser extent the reservoir capacities that are opened up in order to spread those high investment costs across more tonnes of CO_2.

Several directions for further research and development are available for *simCCS*. *SimCCS* currently estimates CO_2 pressure only at the source, assuming that the location and cost of pressure-boosting stations are built into the pipeline parameters. Alternatively, an expanded *simCCS* model could optimally decide where these booster stations should be located, or, alternatively, could simply incorporate the booster stations in the fixed and variable costs of each pipeline arc depending on the length of the arc. Either of these approaches would avoid the oversimplification of assuming that all pressurization occurs at the sources, and would estimate cost subtotals more accurately and possibly change the optimal solution.

This version of *simCCS* is formulated to minimize the costs of meeting a given target for total CO_2 captured and stored. The model could be reformulated to address different economic and policy questions. For instance, the model could be revised to determine endogenously how much CO_2 to capture given carbon taxes or permit costs and potential revenues from selling CO_2 to oil companies for enhanced oil recovery. In such a model, CO_2 would be captured up to the point at which it becomes cheaper to pay the carbon emission tax or buy carbon permits from other emitters.

SimCCS is atemporal, in the sense that all infrastructure components are assumed to be built simultaneously. For example, the 30 $MtCO_2$/yr solution did not build on the 10 and 20 $MtCO_2$/yr solutions. In other words, each new solution on the cost curves in Fig. 5 were reoptimized from a blank slate. The model can be reformulated to optimally select *when*, as well as where and how, to *phase in* CCS infrastructure in stages; the resulting model will be extremely computationally intensive to solve. Thus, an additional area of future emphasis should be on efficient heuristic algorithms to solve the model.

Lastly, *simCCS* can be used to examine the robustness (ability to handle changing CO_2 supplies, capacities, and costs) and resiliency (ability to cope with disruptions and catastrophic failures) of CCS networks.

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Appendix. Equations used for estimates in Section 4

All variables and dimensions are consistent across the equations unless otherwise noted.

Pressurization (Fox and McDonald, 1992; White, 1994):

$$\mathrm{MW} = \frac{\dot{m}\Delta P_d}{\rho\eta} \tag{15}$$

where MW = megawatts, $\dot{m}$ = mass flow rate (kg/s), ΔP_d = pressure drop (Mpa), ρ = fluid density (kg/m^3), and η = pump efficiency.

Pressure drop (Fox and McDonald, 1992; White, 1994):

$$\Delta P_d = f\frac{(\dot{m})^2}{\rho}\frac{L}{D^5}\frac{8000}{\pi^2} \tag{16}$$

where f = pipe friction factor (0.015), L = pipe length (km), and D = pipe diameter (m).

Reservoir injectivity (integrated radial version of Nield and Bejan, 1998)[5]:

$$\dot{m} = \frac{2\pi h}{\ln(r_1/r_2)}\frac{\kappa\rho}{\mu}(P_2 - P_1) \tag{17}$$

where $\ln(r_1/r_2) \approx 7.5$ (McCoy and Rubin, 2005), h = formation thickness (m), k = permeability (m^2), u = fluid viscosity (MPa-s), r = radius (m), P_2 and P_1 = injection and farfield pressures (MPa), respectively.

Gravitational head gain:

$$\Delta P_h = 1000\rho g d \tag{18}$$

where g = gravitational acceleration (9.81 m/s^2) and d = height of fluid column, depth into subsurface (km).

CO_2 density:

$$\begin{aligned}\rho = {} & 748.2 - 5.342(T - T_0) + 12.81(P - P_0) + \ldots \\ & + 0.02414(T - T_0)^2 - 0.6774(P - P_0)^2 + \ldots \\ & + 0.1924(T - T_0)(P - P_0) + 1.585 \times 10^{-4}(T - T_0)^3 + \ldots \\ & + 0.02985(P - P_0)^3 - 4.342 \times 10^{-3}(T - T_0)^2(P - P_0) - \ldots \\ & - 2.740 \times 10^{-3}(T - T_0)(P - P_0)^2\end{aligned} \tag{19}$$

where T = temperature, $T_0 = 340$ K, P = pressure, and $P_0 = 25$ MPa.

CO_2 viscosity:

$$\begin{aligned}\mu = {} & 0.06370 - 7.087 \times 10^{-4}(T - T_0) + 1.797 \times 10^{-3}(P - P_0) + \ldots \\ & + 7.240 \times 10^{-6}(T - T_0)^2 - 4.163 \times 10^{-5}(P - P_0)^2 - \ldots \\ & - 1.993 \times 10^{-6}(T - T_0)(P - P_0) + 4.007 \times 10^{-8}(T - T_0)^3 + \ldots \\ & + 1.240 \times 10^{-6}(P - P_0)^3 - 3.593 \times 10^{-7}(T - T_0)^2(P - P_0) + \ldots \\ & + 5.896 \times 10^{-7}(T - T_0)(P - P_0)^2\end{aligned} \tag{20}$$

where u = viscosity (mPa-s).[6]

Well drilling cost:

$$\begin{aligned}\$ = {} & 38,450\, d^4 - 321,900\, d^3 + 915,400\, d^2 \\ & - 530,100\, d + 161,550\end{aligned} \tag{21}$$

Well equipping cost:

$$\$ = 7,002 + 90,220\ \mathrm{d} \tag{22}$$

Well drilling rig cost:

$$\$ = d(2200\, C_{bbl} + 183,100) \tag{23}$$

C_{bbl} = price ($) of a barrel of oil.

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[5] $V = -\frac{\kappa}{\mu}\frac{\partial P}{\partial x}$(Nield and Bejan, 1998), where V = average fluid velocity, $V = \frac{\dot{m}}{\rho A}$, and A = area$\Rightarrow \frac{\dot{m}}{\rho}\frac{1}{2\pi r h} = \frac{\kappa}{\mu}\frac{dP}{dr} \Rightarrow \frac{\dot{m}}{2\pi h}\int_{r_1}^{r_2}\frac{r}{dr} = \frac{\kappa\rho}{\mu}\int_{P_1}^{P_2} dP$

[6] Note that the dimensions on the viscosity calculation in (20) are (mPa-s), whereas they are (MPa-s) in the injectivity calculation in (17).

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